Three-Way Word Embeddings for Polingual: Script, Sound and Meaning, and a Re-Ranked Lexicon

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Abstract

Polingual translates single words offline from a Wiktionary lexicon for 28 languages. The lexicon lists English meanings in English frequency order, so a rare sense can come first: Hebrew *akhil* (edible) was the first translation of English *good*. We embed 529,131 words of the 29 languages four ways: rendered script pixels with a small CNN, character n-grams, phonetic features from Wiktionary IPA with a grapheme model for words without IPA, and LaBSE meaning vectors, plus a joint projection of all four [empirical: dataset v1/manifest.json, 2026-11-06, python3 apple/scripts/embed/store.py]. Each view is scored on its own task. The vectors, models and pair scores are stored as dataset v1 at `~/agfarms/data/polingual-embeddings/v1` on the build machine. Meaning vectors then re-rank every translation pair, with gloss senses added as candidates. On an independent Tatoeba word set the app's word precision rises from 67.5% to 77.2% [empirical: data/swift_before.txt and data/swift_after.txt, 2026-11-06, TRANSLATE_EVAL=current swift test –filter translateEvalReport], and the first English choice for ten polysemous words is right in 115 of 158 language cases, up from 81 [empirical: dataset v1/reports/regression_report.json, 2026-11-06, python3 apple/scripts/embed/regression.py]. The shipped change is the re-ranked lexicon, 0.24 MB larger.

1 Introduction

A word lookup in Polingual returns up to three English senses from the bundled lexicon. Precision on the Wiktionary reference row was 98.7% for the lexicon of pull request 1198, while precision on a word set aligned from Tatoeba sentence pairs was 67.5% [empirical: data/swift_before.txt and data/swift_after.txt, 2026-11-06, TRANSLATE_EVAL=current swift test –filter translateEvalReport]. The gap comes from sense order: Wiktionary translation tables list a foreign word under every English headword that uses it, and the builder kept the four most frequent English headwords. A foreign word that translates a rare sense of a frequent English word inherits that word.

The founder asked for three separate embeddings per word, of script, sound and meaning, and a joint one, and for higher accuracy. This paper builds the four, measures each against a task it should solve, and uses them to score every translation pair in the lexicon.

2 Related work

LaBSE trains a dual encoder on translation pairs for 109 languages and places translations near each other [1]. Bilingual lexicon induction retrieves translations by nearest neighbour across aligned spaces [2]. Subword n-grams give vectors to unseen spellings [3]. Rendered text as model input handles noise and unseen scripts [4, 5]. Contrastive objectives with in-batch negatives train both

our script encoder and our joint projection [6, 7]. Tatoeba sentence pairs are a standard multilingual test source [8]. Phonetic features come from the PanPhon feature table and IPA from WikiPron, a scrape of Wiktionary pronunciations; neither has a DOI or arXiv identifier, so we name them with their repository paths in 4.

3 Preliminaries

Definition 3.1 (Lexicon)**.** For a language L, the lexicon maps a headword w to an ordered list $E(w)$ of English words, at most four of which ship, and a gloss $g(w)$, a list of short English sense strings from the word's own Wiktionary entry. The app shows the first three items of $E(w)$, or $g(w)$ when $E(w)$ is empty. For English word e the lexicon also holds an ordered list $F(e)$ of target words; English-to-target lookups show the first.

Definition 3.2 (Modality)**.** A modality m is a map from words to unit vectors in $\mathbb{R}^{d_m}$: script $d = 64$, character $d = 64$, phonetic $d = 96$, meaning $d = 768$, joint $d = 256$. The similarity of words a, b under m is the dot product $\cos_m(a, b)$ of their unit vectors.

Definition 3.3 (Joint score)**.** Given weights w_m, the joint score of a pair is $s(a, b) = \sum_m w_m \cos_m(a, b)$. With weight one on meaning and zero elsewhere it equals the meaning cosine (Lean: `jointScore_meaning_only`).

Definition 3.4 (Precision)**.** For an evaluation row with J judged answers of which R are right, precision is $100R/J$ percent; it never exceeds 100 when $R \le J$ (Lean: `precisionPct_le`).

4 Method

Vocabulary. The lexicon builder of pull request 1198 runs in a dump mode that writes, for every shipped headword, all English candidates in frequency order and the gloss, and for every English headword all target candidates. Gloss senses that equal an English headword add target candidates to its list. The vocabulary is every headword, every candidate and every evaluation word: 26,887 English words and 502,244 words across the 28 target languages [empirical: dataset v1/manifest.json, 2026-11-06, python3 apple/scripts/embed/vocab.py].

Script. Each word is rendered in Noto Sans for Latin, Greek and Cyrillic, Noto Sans Arabic, Hebrew, Devanagari, Tamil and Thai, and Noto Sans CJK for Chinese, Japanese and Korean, with HarfBuzz shaping through Pillow's Raqm layout, into a 32×128 grey raster. A four-block CNN maps the raster to 64 dimensions. It trains for 2,000 steps on 116,000 words rendered at font weights 300, 400 and 700; two views of a word, each with random scale, shear, shift and noise, form a positive pair under a symmetric InfoNCE loss with temperature 0.07. The character view hashes the word's 1 to 3-grams with boundary marks into 2^{20} buckets and sums a fixed Gaussian vector per bucket.

Phonetic. WikiPron files `CUNY-CL/wikipron/data/scrape/tsv` give space-separated IPA; we use up to 150,000 entries per language, 1,589,580 in all [empirical: dataset v1/reports/phonetic_report.json, 2026-11-06, python3 apple/scripts/embed/phonetic.py]. Each segment maps to the 24 PanPhon features (`dmort27/panphon`) in $\{-1, 0, 1\}$; the word vector averages the segment features inside four equal position bins, giving 96 dimensions that keep coarse order. For words without IPA a grapheme model predicts the vector: the word splits into units, extended grapheme clusters by default, Hangul jamo for Korean, pinyin initials and finals for Chinese, kana with small-kana merging for Japanese; each

unit, each unit at its position bin, each unit bigram and the first and last unit index a learned 96-d vector in an EmbeddingBag, and the mean of those vectors regresses the IPA vector under squared error. The unit vectors are the letter and grapheme-cluster level of the phonetic embedding. 313,678 of the 529,131 vocabulary rows carry IPA; the grapheme model fills the rest [empirical: dataset v1/reports/phonetic_report.json, 2026-11-06, python3 apple/scripts/embed/phonetic.py].

Meaning. LaBSE encodes each bare word [1]. Gloss strings are encoded the same way, with parenthesised notes removed.

Joint. Two joint forms are fitted on translation pairs from 80% of each language's headwords, split by hash [empirical: dataset v1/reports/joint_report.json, 2026-11-06, python3 apple/scripts/embed/joint.py]; the other 20% are test queries [empirical: dataset v1/reports/joint_report.json, 2026-11-06, python3 apple/scripts/embed/joint.py]. The weighted form searches $w_m \in \{0, 0.05, 0.1, 0.2, 0.3\}$ for script, character and phonetic with meaning fixed at one, maximising P@1 on 300 training queries per language; it selects 0.05 for script and zero for the other two [empirical: dataset v1/reports/joint_report.json, 2026-11-06, python3 apple/scripts/embed/joint.py]. A logistic regression on the four cosines gave a phonetic weight of -2.05, which reflects that sound similarity between a foreign word and its English translation is noise outside cognates, and was dropped. The projection form scales each modality, concatenates the 992 dimensions and maps them linearly to 256, trained for 4,000 steps of symmetric InfoNCE at temperature 0.05 with duplicate English targets masked.

Re-ranking. For headword w, candidates are $E(w)$ in frequency order j plus up to three gloss senses at gloss position j. Candidate x scores $\cos(w, x) - \beta \log(1+j)$, plus a bonus b for the first gloss sense, and the top four ship. For English e, target candidate t scores $\cos(e, t) - \beta_r \log(1+j) + \gamma\, \pi(t)$ with $\pi(t) = 1 - \log(1 + r_t)/\log(60001)$ from the wordfreq rank r_t of t in its language, zero when unranked, and a candidate spelled the same as e drops to the end. The scorer, β, a drop threshold, a relative cut, gloss use and b were chosen on the even rows of the Tatoeba-aligned word set by first-sense precision with coverage held at the baseline; β_r and γ on the even half of 378 common English words with Tatoeba-derived targets. Selected: meaning cosine, $\beta = 0.05$, no drop threshold, gloss on, $b = 0.1$, $\beta_r = 0.05$, $\gamma = 0.5$ [empirical: dataset v1/reports/rerank_report.json, 2026-11-06, python3 apple/scripts/embed/rerank.py]. The odd rows and odd words are held out.

English-to-target references. For each English word and language, target tokens are scored by Dice co-occurrence with the English token over Tatoeba pairs, character 1 to 3-grams for Chinese, Japanese and Thai, and the top three with count at least three are the references. A first choice is right when it equals a reference, shares its first four letters with one, or for the unspaced languages contains or is contained in one.

5 Results

5.1 Each embedding on its own task

The script CNN finds the listed confusable of a letter in its top ten for 55.2% of pairs against 48.5% for raw pixel cosine [empirical: dataset v1/reports/script_report.json, 2026-11-06, python3 apple/scripts/embed/script_eval.py]; character n-grams cannot, since distinct code points share no n-gram. For fuzzy search, one random edit per word, character n-grams recover the word at rank one in 73.7% of cases

Table 1: Unicode confusable pairs, 924 single letters, ranked among a pool of 4,996 letters [empirical: dataset v1/reports/script_report.json, 2026-11-06, python3 apple/scripts/embed/script_eval.py].

View	R@1	R@10	MRR
Script CNN	37.2	55.2	0.432
Raw pixels	34.8	48.5	0.397
Character n-grams	0.2	0.2	0.003

on average over 12 languages and the CNN in 52.8% [empirical: dataset v1/reports/script_report.json, 2026-11-06, python3 apple/scripts/embed/script_eval.py]; their sum reaches 77.2% [empirical: dataset v1/reports/script_report.json, 2026-11-06, python3 apple/scripts/embed/script_eval.py]. For a swap to a visual confusable the CNN reaches 85.1% and the sum 93.6%, against 77.6% for n-grams [empirical: dataset v1/reports/script_report.json, 2026-11-06, python3 apple/scripts/embed/script_eval.py]. 2 lists each language.

Table 2: Fuzzy search R@1 (%) within each language's vocabulary [empirical: dataset v1/reports/script_report.json, 2026-11-06, python3 apple/scripts/embed/script_eval.py].

	one random edit			confusable swap		
Lang	CNN	n-gram	sum	CNN	n-gram	sum
es	59.6	72.4	77.4	92.9	75.9	95.8
de	67.1	77.8	82.0	96.0	79.5	98.1
ru	67.1	81.1	85.7	99.4	83.5	99.2
el	64.5	85.3	88.5	97.8	85.1	98.7
ar	28.4	55.8	55.5	44.4	59.2	71.8
he	28.2	57.0	57.7	65.2	63.4	87.2
hi	32.3	67.9	64.6	100.0	96.4	100.0
ta	51.5	77.7	75.2	69.3	81.4	91.1
th	46.6	76.2	77.7	93.3	74.6	94.0
ja	49.3	70.3	81.0	92.0	70.0	96.2
zh	87.9	90.6	97.6	85.4	84.5	97.1
ko	50.9	72.4	83.4	–	–	–

The phonetic view finds homophones better than character n-grams in 23 of 27 languages with enough pairs, mean R@10 80.2% against 62.3% [empirical: dataset v1/reports/phonetic_report.json, 2026-11-06, python3 apple/scripts/embed/phonetic.py]. The gain is largest where spelling hides sound: Chinese 81.3% against 6.5%, Korean 87.8% against 7.6%, Japanese 33.7% against 5.6% [empirical: dataset v1/reports/phonetic_report.json, 2026-11-06, python3 apple/scripts/embed/phonetic.py].

Meaning carries translation: P@10 57.1% on average, against 10.8% for script, 11.3% for character and 5.6% for phonetic [empirical: dataset v1/reports/joint_report.json, 2026-11-06, python3 apple/scripts/embed/joint.py]. The joint projection reaches 58.9% P@10 and 36.0% P@1, against 37.1% P@1 for meaning alone [empirical: dataset v1/reports/joint_report.json, 2026-11-06, python3 apple/scripts/embed/joint.py]; the weighted form matches meaning within 0.1 points. Across related language pairs (5) the phonetic view is the one form-based view that crosses scripts: Russian to Polish R@10 9.7% against 0.3% for character n-grams [empirical: dataset v1/reports/joint_report.json, 2026-11-06, python3 apple/scripts/embed/joint.py].

Table 3: Phonetic view. G2P: the grapheme model on held-out WikiPron words, mean cosine to the IPA vector and R@10 among 3,000. Homophones: distinct spellings with identical IPA, query and pool both predicted from spelling, R@10 (%) for phonetic and character views [empirical: dataset v1/reports/phonetic_report.json, 2026-11-06, python3 apple/scripts/embed/phonetic.py].

Lang	WikiPron	IPA rows	G2P cos	G2P R@10	pairs		phon	char
en	79306	8198/26887	0.913	57.4	1000	76.5	55.8	
es	131056	15650/22864	0.965	94.8	1000	96.3	85.6	
fr	84402	14929/20813	0.918	65.8	1000	93.8	87.2	
it	81441	14414/19915	0.957	93.1	302	84.4	77.8	
pt	56305	12806/18250	0.944	80.2	977	85.6	71.8	
de	51318	10219/20538	0.923	60.3	671	86.9	75.9	
nl	45234	11852/19425	0.927	61.9	779	84.5	75.2	
sv	5054	3060/22446	0.828	5.5	72	81.9	63.9	
pl	146081	16759/18832	0.949	94.8	1000	92.9	76.3	
fi	150000	14490/21587	0.959	93.0	216	72.2	75.5	
ru	149999	5675/17355	0.954	89.9	1000	99.5	97.3	
el	19008	6367/14964	0.934	57.9	1000	97.1	71.4	
ar	14272	6952/14265	0.908	74.6	97	48.5	76.3	
hi	30379	12189/14862	0.909	66.0	832	73.7	85.5	
ja	44247	11607/18346	0.867	18.4	1000	33.7	5.6	
zh	147316	8052/19348	0.886	72.4	1000	81.3	6.5	
ko	30421	16320/20247	0.927	79.3	370	87.8	7.6	
he	2872	2782/15001	0.821	13.5	45	88.9	42.2	
fa	10728	10253/17162	0.862	31.6	117	85.5	53.8	
tr	11436	8007/20509	0.889	23.1	–	–	–	
cs	63095	21081/22952	0.950	97.3	849	95.5	76.2	
id	17810	11381/17605	0.916	45.9	47	66.0	57.4	
vi	26978	11920/23449	0.893	62.6	527	53.7	23.1	
th	17221	13457/14174	0.845	34.8	589	63.7	55.9	
ta	10095	9345/10873	0.906	57.0	–	–	–	
la	41656	1080/16014	0.946	90.0	496	95.6	90.1	
grc	57540	11293/13020	0.946	80.7	1000	94.4	53.6	
sa	9796	8796/10361	0.894	37.1	30	53.3	56.7	
ang	54514	14744/17067	0.951	95.0	675	93.2	78.8	

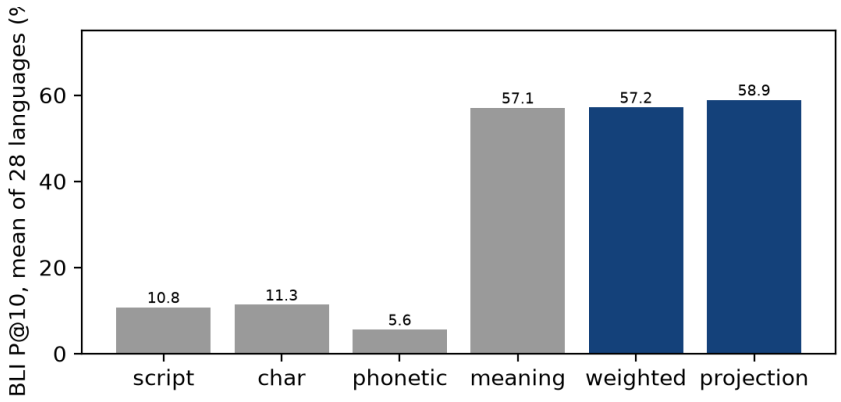


Figure 1: Bilingual lexicon induction, target word to English among 26,887 English words, P@10 averaged over 28 languages, test headwords only [empirical: dataset v1/reports/joint_report.json, 2026-11-06, python3 apple/scripts/embed/joint.py].

Table 4: Bilingual lexicon induction P@10 (%) per language and view; gold is the Wiktionary candidate set [empirical: dataset v1/reports/joint_report.json, 2026-11-06, python3 apple/scripts/embed/joint.py].

Lang	n	script	char	phon	meaning	weighted	projection
es	506	26.3	28.5	12.1	68.4	68.4	71.3
fr	540	35.0	38.3	14.1	72.2	72.4	74.3
it	479	25.3	28.6	10.6	71.6	71.8	74.1
pt	454	31.3	31.5	9.5	74.2	74.4	77.5
de	456	20.6	24.3	10.5	65.4	65.6	67.1
nl	477	27.9	27.7	15.5	72.3	72.1	75.7
sv	428	25.5	25.5	5.6	64.0	63.8	64.7
pl	391	16.6	16.9	10.7	76.0	76.5	77.0
fi	542	7.4	8.3	2.4	55.5	55.7	55.7
ru	416	3.1	0.2	4.1	72.6	71.9	73.3
el	698	1.3	0.1	2.6	59.0	59.5	61.7
ar	758	0.1	0.1	0.0	17.8	17.7	23.6
hi	630	0.2	0.2	6.2	59.5	59.2	61.3
ja	359	0.0	0.3	4.5	72.7	72.1	73.5
zh	156	0.0	0.0	1.3	67.9	67.3	67.9
ko	670	0.0	0.0	8.5	61.8	62.2	62.8
he	460	0.0	0.2	0.4	37.0	36.5	38.7
fa	449	0.2	0.2	3.3	40.8	41.2	44.5
tr	681	11.0	12.0	3.1	54.6	54.9	55.2
cs	703	17.8	16.6	7.7	63.0	63.2	67.1
id	421	21.1	23.0	6.9	69.1	69.1	71.0
vi	498	6.6	4.4	0.4	59.2	60.0	60.4
th	418	0.2	0.2	4.3	61.2	62.2	59.6
ta	213	0.0	0.0	0.9	61.0	61.0	63.4
la	510	16.1	18.8	1.2	40.4	41.2	41.8
grc	284	0.0	0.0	2.8	25.0	25.4	26.8
sa	136	0.0	0.0	0.0	37.5	37.5	40.4
ang	148	7.4	10.8	6.8	17.6	17.6	18.2
mean		10.8	11.3	5.6	57.1	57.2	58.9

Table 5: Translation retrieval between related languages, R@10 (%) [empirical: dataset v1/reports/joint_report.json, 2026-11-06, python3 apple/scripts/embed/joint.py].

Pair	n	script	char	phon	meaning	weighted	projection
es-pt	417	56.4	55.2	38.8	78.4	78.7	77.0
es-it	411	41.1	36.7	36.7	69.1	70.8	71.5
fr-it	402	31.8	30.3	12.4	68.4	69.7	68.9
de-nl	341	30.5	35.2	27.9	68.3	69.8	67.7
sv-de	350	28.6	31.1	8.6	58.3	60.6	60.3
cs-pl	355	33.5	23.4	25.9	71.0	71.0	69.6
ru-pl	298	3.7	0.3	9.7	76.8	75.8	75.8
hi-sa	257	20.2	26.1	13.6	41.6	40.9	41.6
ar-fa	433	19.2	8.3	3.7	23.1	24.0	27.7
ja-zh	133	30.8	31.6	0.0	66.9	66.9	65.4
el-grc	255	36.1	33.7	19.6	42.4	42.7	45.5
la-it	353	22.1	22.7	2.5	39.9	40.2	41.6

5.2 Re-ranked lexicon

Table 6: Held-out odd rows of the Tatoeba-aligned word set, 2,325 words, Python replica of the app lookup [empirical: dataset v1/reports/rerank_report.json, 2026-11-06, python3 apple/scripts/embed/rerank.py].

Precision (%)	before	after
Any shown sense	67.3	78.5
First shown sense	58.0	68.3

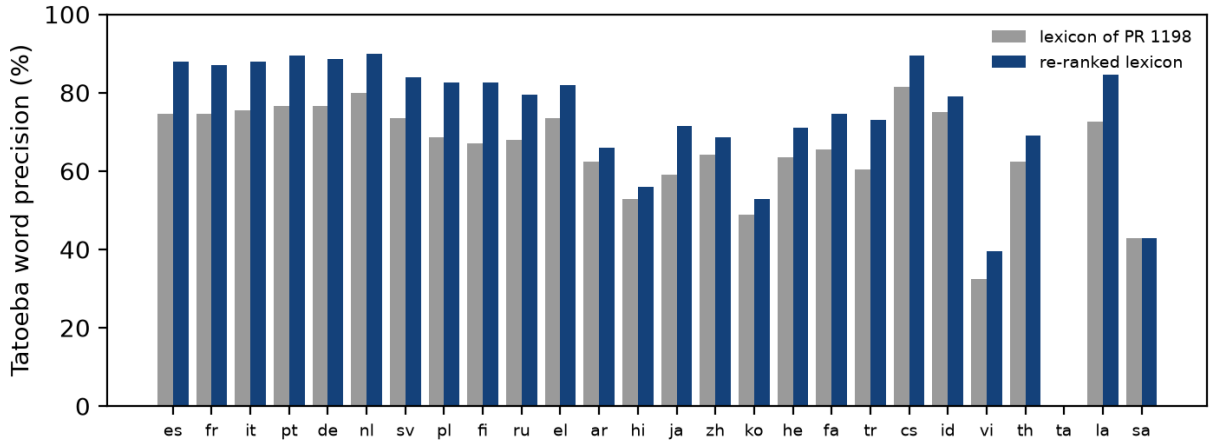


Figure 2: Tatoeba-aligned word precision per language in the app evaluation [empirical: data/swift_before.txt and data/swift_after.txt, 2026-11-06, TRANSLATE_EVAL=current swift test –filter translateEvalReport].

Phrase precision, a word-by-word gloss scored by token F1, rises from 14.1% to 17.2% and sentence precision from 7.7% to 9.9% [empirical: data/swift_before.txt and data/swift_after.txt, 2026-11-06, TRANSLATE_EVAL=current swift test –filter translateEvalReport]; the Wiktionary reference row stays at 98.7% [empirical: data/swift_before.txt and data/swift_after.txt, 2026-11-06, TRANSLATE_EVAL=current swift test –filter translateEvalReport]. On the held-out half, precision with any shown sense rises from 67.3% to 78.5%, and with the first shown sense from 58.0% to 68.3% [empirical: dataset v1/reports/rerank_report.json, 2026-11-06, python3 apple/scripts/embed/rerank.py]. In the app evaluation, all rows, Tatoeba word precision rises from 67.5% to 77.2% [empirical: data/swift_before.txt and data/swift_after.txt, 2026-11-06, TRANSLATE_EVAL=current swift test –filter translateEvalReport]. For English to target, the first choice for ten polysemous words (good, right, light, fair, well, kind, mean, bank, play, run) is right in 115 of 158 language cases, up from 81; English *good* to Hebrew now gives *tov* in place of *akhil* [empirical: dataset v1/reports/regression_report.json, 2026-11-06, python3 apple/scripts/embed/regression.py]. On the held-out half of common English words, first-choice accuracy rises from 60.6% to 68.4% [empirical: dataset v1/reports/rerank_report.json, 2026-11-06, python3 apple/scripts/embed/rerank.py].

6 Calibration

Every hyperparameter was chosen on one half of an evaluation set and is reported on the other half (6, 8). The Python replica of the app lookup gave 66.8% on the full Tatoeba row for the

Table 7: App evaluation, `TranslateEvalTests`, target word to English. Tatoeba: word precision on the Tatoeba-aligned row, all rows. Wiktionary: precision on the Wiktionary reference row [empirical: data/swift_before.txt and data/swift_after.txt, 2026-11-06, TRANSLATE_EVAL=current swift test –filter translateEvalReport].

	Tatoeba word precision (%)			Wiktionary row (%)	
Lang	before	after	change	before	after
es	74.5	88.0	+13.5	99.6	99.6
fr	74.5	87.0	+12.5	99.5	99.5
it	75.5	88.0	+12.5	99.0	99.0
pt	76.5	89.5	+13.0	99.5	99.5
de	76.5	88.5	+12.0	98.9	98.9
nl	80.0	90.0	+10.0	99.7	99.7
sv	73.5	84.0	+10.5	98.4	98.5
pl	68.5	82.5	+14.0	98.5	98.5
fi	67.0	82.5	+15.5	99.4	99.4
ru	68.0	79.5	+11.5	99.9	99.9
el	73.5	82.0	+8.5	99.2	99.2
ar	62.5	66.0	+3.5	100.0	100.0
hi	52.9	55.9	+3.0	99.3	99.3
ja	59.0	71.5	+12.5	100.0	100.0
zh	64.1	68.7	+4.6	99.8	99.8
ko	48.9	52.8	+3.9	99.3	99.4
he	63.5	71.0	+7.5	99.4	99.6
fa	65.5	74.5	+9.0	99.9	99.9
tr	60.5	73.0	+12.5	99.5	99.5
cs	81.5	89.5	+8.0	99.8	99.8
id	75.0	79.0	+4.0	99.3	99.3
vi	32.5	39.5	+7.0	81.5	81.5
th	62.5	69.0	+6.5	98.7	98.7
ta	0.0	0.0	+0.0	99.5	99.5
la	72.5	84.5	+12.0	99.7	99.7
sa	42.9	42.9	+0.0	98.9	98.9
all	67.5	77.2	+9.7	98.7	98.7

Table 8: English to target, first choice right (%) against Tatoeba-derived references [empirical: dataset v1/reports/regression_report.json, 2026-11-06, python3 apple/scripts/embed/regression.py].

Set	cases	before	after
Polysemous set, 10 words	158	51.3	72.8
Common English words, held-out half	2025	60.6	68.4

lexicon of pull request 1198, where the Swift evaluation gave 67.5% [empirical: data/swift_before.txt and data/swift_after.txt, 2026-11-06, TRANSLATE_EVAL=current swift test –filter translateEvalReport]; the replica skips the deck corpus layer, clitic stripping and NLTagger lemmas, which accounts for the 0.7 point gap. The re-ranked lexicon builds from the same inputs as the shipped one: a rebuild of pull request 1198's lexicon with this pipeline in dump mode reproduces every line of every file. The Tatoeba references are noisy: they are the three English words that co-occur most with the target word, so a right sense that is less frequent in Tatoeba counts as wrong. The error budget is the sampling error of a proportion near 0.7 over 2,325 held-out words, a standard error of 0.95 points, small next to the 11.2 point change.

7 Limitations

Longer answers help the any-sense metric: the mean answer grew from 5.35 to 8.58 words, because gloss senses are phrases, which is why the first-sense metric is reported beside it. The English-to-target Wiktionary row falls from 94.7% to 89.1% [empirical: data/swift_before.txt and data/swift_after.txt, 2026-11-06, TRANSLATE_EVAL=current swift test –filter translateEvalReport]: its references are the translation-table lists themselves, and target words added from gloss senses, such as Japanese *ii* for *good*, count as wrong there while the Tatoeba-derived references in 8 count them right. LaBSE has no Latin, Ancient Greek, Sanskrit or Old English training data, so meaning scores there lean on shared subwords. The phonetic view ignores tone, so Mandarin, Thai and Vietnamese homophones that differ by tone collapse. Swedish and Hebrew have about 5,000 and 2,900 WikiPron entries, too few for a good grapheme model: G2P R@10 is 5.5% and 13.5% [empirical: dataset v1/reports/phonetic_report.json, 2026-11-06, python3 apple/scripts/embed/phonetic.py]. The script CNN trains on weight changes of one font family per script, so it learns little about handwriting or other typefaces. Tamil has no usable Tatoeba-aligned references, since the alignment tokenizer splits Tamil vowel signs. The joint projection improves P@10 but not P@1 over meaning, and the re-ranker selected the meaning cosine; the joint vectors ship in the dataset and not in the app.

Licenses. Wiktionary data through kaikki.org and WikiPron is CC BY-SA 4.0, and the re-ranked lexicon and dataset carry it. Tatoeba sentences are CC BY 2.0 FR. Noto fonts are SIL OFL 1.1. LaBSE is Apache 2.0. PanPhon is MIT.

References

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A Glossary

lexicon https://github.com/bucket-foundation/bucket-foundation/blob/feat/apple-translate-embed/apple/scripts/translate/build_lexicon.py
The per-language Polingual word list: each headword with up to four English meanings, a gloss, a romanization, form links and English-to-target lists; see 3.1.

modality https://github.com/bucket-foundation/bucket-foundation/blob/feat/apple-translate-embed/apple/scripts/embed/common.py
One of the four word views embedded here: script, character, phonetic, meaning; see 3.2.

joint score https://github.com/bucket-foundation/bucket-foundation/blob/feat/apple-translate-embed/apple/scripts/embed/joint.py
A weighted sum of per-modality cosines, or the cosine of a learned projection of all four vectors; see 3.3.

precision https://github.com/bucket-foundation/bucket-foundation/blob/feat/apple-translate-embed/apple/BucketKit/Tests/BucketKitTests/TranslateEvalTests.swift
Judged answers marked right divided by judged answers; see 3.4.

B Lean listings

`lean/Bucket/Embed.lean` states the dot product, the joint score, precision in whole percent and sense filtering, and proves that the meaning-only joint score is the meaning cosine, that precision is at most 100 when right answers do not exceed judged ones, and that filtering senses never lengthens a list.

```
def jointScore (w c : List Int) : Int := dot w c
theorem jointScore_meaning_only (s p c m : Int) :
    jointScore [0, 0, 0, 1] [s, p, c, m] = m
def precisionPct (right judged : Nat) : Nat := right * 100 / judged
theorem precisionPct_le (right judged : Nat) (h : right <= judged) :
    precisionPct right judged <= 100
theorem keepSenses_length (keep : a -> Bool) (senses : List a) :
    (keepSenses keep senses).length <= senses.length
```

`lake build` output, captured by `make lean`. This block is a verbatim transcript.

```
info: polingual_embeddings: no previous manifest, creating one from scratch
info: toolchain not updated; already up-to-date
Built Bucket.Embed (1.0s)
Built Bucket (157ms)
Build completed successfully (4 jobs).
```